Egs

6. S.T $\displaystyle\int_0^\infty \frac{e^{-x^3}}{\sqrt{x}}\,dx \int_0^\infty y^4 \cdot e^{-y^6}\,dy = \frac{\pi}{9}$.

Sol^n: Let $I_1 = \displaystyle\int_0^\infty e^{-x^3} \cdot x^{-1/2}\,dx$ & $I_2 = \displaystyle\int_0^\infty y^4 e^{-y^6}\,dy$

In I_1, put $x^3 = t$

$\Rightarrow x = t^{1/3}$

$\Rightarrow dx = \frac{1}{3} t^{-2/3}\,dt$

In I_2 put $y^6 = u$

$\Rightarrow y = u^{1/6}$

$\Rightarrow dy = \frac{1}{6} u^{-5/6}\,du$

x	0	∞
$t = x^3$	0	∞

$\therefore I_1 = \displaystyle\int_0^\infty e^{-t} \left(t^{1/3}\right)^{-1/2} \cdot \frac{1}{3} t^{-2/3}\,dt$; $I_2 = \displaystyle\int_0^\infty e^{-t} \left(u^{1/6}\right)^4 \cdot \frac{1}{6} u^{-5/6}\,du$

$\therefore I_1 \times I_2$

$= \frac{1}{3} \displaystyle\int_0^\infty e^{-t} t^{-1/6} \cdot t^{-2/3}\,dt \times \frac{1}{6} \int_0^\infty e^{-t} u^{4/6} \cdot u^{-5/6}\,du$

$= \frac{1}{18} \displaystyle\int_0^\infty \underline{e^{-t}\, t^{-5/6}}\,dt \cdot \int_0^\infty e^{-t} u^{-1/6}\,du$

$= \frac{1}{18} \Gamma\left(\frac{1}{6}\right) \cdot \Gamma\left(\frac{5}{6}\right)$

$= \frac{1}{18} \cdot \frac{\pi}{\sin \pi \cdot \frac{1}{6}}$ $\left(\because \Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}\right)$

$= \frac{1}{18} \cdot \frac{\pi \times 2}{1} = \frac{\pi}{9}$ //

7. S.T $\int_0^1 x^m (\log x)^n \, dx = \frac{(-1)^n}{(m+1)^{n+1}} \Gamma(n+1)$

$\int_0^\infty e^{-x} x^{k-1} \, dx$

$\log x = t$

$\log 0 = -\infty$

$\log 1 = t$

$\Rightarrow e^t = 1$

$\Rightarrow t = 0$

$\log 0 = t$

$\Rightarrow e^t = 0$

$t = -\infty$

$e^{-x} = \frac{1}{e^x}$

$e^{-0.2} = t$ 0.0001

If $x = e^t$

$x = 1 \Rightarrow t = 0$

$x = 0, t = -\infty$

Soln: put $\log x = -t$

$\Rightarrow e^{-t} = x$

$\Rightarrow x = e^{-t}$

$\Rightarrow dx = -e^{-t} \, dt$

x	0	1
$t = -\log x$	$-(-\infty)$	0
	∞	0

$\therefore \int_0^1 x^m (\log x)^n \, dx$

$= \int_\infty^0 (e^{-t})^m (-t)^n (-e^{-t} \, dt)$

$\left[\because \int_a^b -dx = \int_b^a dx \right]$

$\Rightarrow \int_0^1 x^m (\log x)^n \, dx$

$= \int_0^\infty e^{-mt} \cdot (-1)^n t^n \cdot e^{-t} \, dt$

$= (-1)^n \int_0^\infty e^{-t(m+1)} \cdot t^n \, dt$

put $t(m+1) = u$

$\Rightarrow t = \frac{u}{m+1}$

$dt = \frac{1}{m+1} du$

t	0	∞
u	0	∞

$\therefore I = (-1)^n \int_0^\infty e^{-u} \cdot \left(\frac{u}{m+1}\right)^n \cdot \frac{1}{m+1} du$

$\Rightarrow I = \frac{(-1)^n}{(m+1)^{n+1}} \int_0^\infty e^{-u} u^n \, du$

$\Rightarrow I = \frac{(-1)^n}{(m+1)^{n+1}} \Gamma(n+1) = \text{RHS}$

8. Evaluate $\int_0^1 \frac{dx}{\sqrt{x\left(\log\frac{1}{x}\right)}}$ $\left[\text{Ans. } \sqrt{2\pi}\right]$

$\log\left(\frac{1}{x}\right) = \log 1 - \log x$
$= 0 - \log x$

9. Ev. $\int_0^1 x^3 \left[\log\left(\frac{1}{x}\right)\right]^4$ $\left[\text{Ans } \frac{1}{4}\right]$

8. Sol^n: Put $\log\left(\frac{1}{x}\right) = -\log x = t$

$\Rightarrow \log x = -t$

$\Rightarrow x = e^{-t}$

$\Rightarrow dx = -e^{-t}\,dt$

x	0	1
$t = -\log x$	$-(-\infty)$	0

$\therefore I = \int_0^1 \frac{dx}{\sqrt{x\left(\log\frac{1}{x}\right)}}$

$= \int_0^1 dx \cdot x^{-1/2}\left(\log\frac{1}{x}\right)^{-1/2}$

$= \int_\infty^0 -e^{-t}\,dt \cdot \left(e^{-t}\right)^{-1/2} \cdot \left(t^{-1/2}\right)$ $\left[\because \int_\infty^0 -dt = \int_0^\infty dt\right]$

$= \int_0^\infty e^{-t\left(1-\frac{1}{2}\right)} t^{-1/2}\,dt$

$= \int_0^\infty e^{-t/2} t^{-1/2}\,dt$

put $\frac{t}{2} = u$

$\Rightarrow t = 2u$

$dt = 2\,du$

t	0	∞
u	0	∞

$\therefore I = \int_0^\infty e^{-u} \cdot (2u)^{-1/2} \cdot 2\,du$

$= 2 \cdot 2^{-1/2} \int_0^\infty e^{-u} u^{-1/2}\,du$

$= 2^{1/2}\,\Gamma\left(\frac{1}{2}\right)$

$= 2^{1/2}\sqrt{\pi}$

$= \sqrt{2\pi}$

9. $\int_0^1 x^3\left(\log\frac{1}{x}\right)^4 dx$

Sol^n: $\log x = -t$

$\Rightarrow \log\frac{1}{x} = \log 1 - \log x$

$= -\log x$

$= t$

ie $\log\frac{1}{x} = t$

$\Rightarrow \frac{1}{x} = e^t$

$\Rightarrow x = \frac{1}{e^t} = e^{-t}$

$\Rightarrow dx = -e^{-t}\,dt$

x	0	1
$t = \log\left(\frac{1}{x}\right)$	∞	0

$\therefore \int_0^1 x^3\left(\log\frac{1}{x}\right)^4 dx$

$= \int_\infty^0 (e^{-t})^3\, t^4 (-e^{-t}\,dt)$

$\because \int_a^b -dt = \int_b^a dt,$

$I = \int_0^\infty e^{-3t}\cdot e^{-t}\, t^4\, dt$

$\Rightarrow I = \int_0^\infty e^{-4t}\, t^4\, dt$

put $4t = u$

$\Rightarrow t = \frac{u}{4}$

$\Rightarrow dt = \frac{1}{4}du$

t	0	∞
$u = 4t$	0	∞

$\therefore I = \int_0^\infty e^{-u}\cdot\left(\frac{u}{4}\right)^4\cdot\frac{1}{4}\,du$

$= \frac{1}{4^5}\int_0^\infty e^{-u}\cdot u^4\, du$

$= \frac{1}{4^5}\cdot\Gamma(5) = \frac{4!}{4^5} = \frac{6}{4^4}$

Type 1

Limits $\int_0^\infty$, $e^{-f(x)}$

Substitution

put $f(x) = t$

Type 2

Limits $\int_0^1$ $\log x$ form

put $\log x = -t$

Type 3 $\int_0^\infty$ a^x form $a^x = e^{-t}$

$$\int_0^\infty a^x x^3 \, dx$$